Supporting Sign Language Video Comprehension through In-Situ Automatic Segmentation and Sign Lookup Tool

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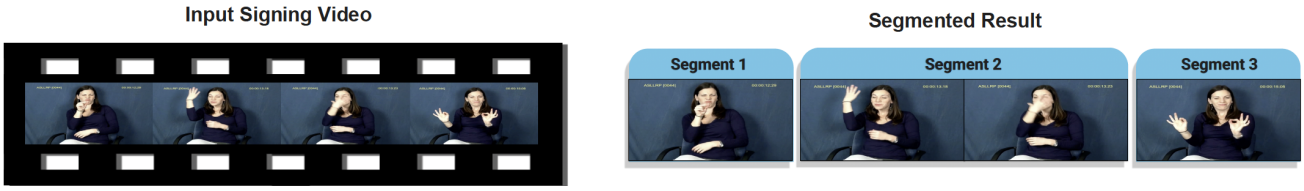


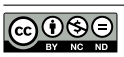
Figure 1: A conceptual overview of how automatic segmentation breaks continuous signing video into sign-level units, helping learners isolate unfamiliar signs, examine them individually, and access candidate matches without leaving the video context.

Abstract

Sign language learners often closely watch signing videos to develop receptive skills. However, they encounter unfamiliar signs, especially early on when they are still developing their vocabulary, and later due to challenges in understanding signs in continuous videos because of coarticulation. Existing tools, such as feature-based dictionaries, often require learners to identify the linguistic properties of a sign to search for it, which can be cumbersome. Prior work has proposed interfaces to support in-situ sign lookup, but end-to-end systems have not yet been disseminated or demonstrated. We demonstrate a fully automatic video comprehension prototype system that combines state-of-the-art sign language segmentation with a fine-tuned spatiotemporal recognition network. The underlying AI pipeline ingests a continuous signing video, applies a linguistically motivated BIO-tagging segmentation framework to locate boundaries, and routes individual segments into a fine-tuned Spatial-Temporal Graph Convolutional Network (ST-GCN) to retrieve the closest gloss matches. We present the system design and demonstration plan. We also discuss the system's current limitations and future research directions.

CCS Concepts

• **Human-centered computing** → *Accessibility technologies*; **User interface programming**.



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Keywords

American Sign Language, Language Learning, Sign Language, Sign Language Learning, Artificial Intelligence, Sign Language Recognition, Sign Language Segmentation, AI Tools

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1 INTRODUCTION

Sign languages, such as American Sign Language (ASL), are widely regarded as one of the most challenging languages to learn [16], in part because they are expressed in a three-dimensional visual-spatial modality and employ grammatical mechanisms that differ fundamentally from those of spoken languages. Signs are produced through combinations of handshape, movement, palm orientation, spatial location, and nonmanual markers (e.g., facial expressions, head movements, and body posture), and their performance within continuous signing utterances can be shaped by the surrounding linguistic context [33].

Videos are commonly used in ASL learning, partly because learners may have limited access to people with whom they can practice, but also because videos are a central component of many ASL curricula [1]. However, a major bottleneck in video-based learning is the limited availability and poor usability of video comprehension tools. Most existing technological resources for ASL learning focus primarily on basic vocabulary and rarely extend beyond helping learners recognize isolated signs in their "citation" form[1]. Existing commercial and research-produced tools are typically designed either as vocabulary-learning games (e.g., [3, 19, 31]) or as

[1]In sign linguistics, a sign in "citation form" refers to its standard or dictionary form, produced in isolation rather than within a sentence or natural discourse [33].

feature-based dictionaries (e.g., [6, 20]) that allow learners to look up unfamiliar signs using their linguistic properties. Yet, identifying signs by linguistic properties can be especially challenging for beginner and intermediate sign language learners [13]. Even if learners attempt to infer these features from context, they must completely leave their video-viewing environment to perform a search, severely disrupting the comprehension process. Currently, no commercially available tools provide fluid, in-situ lookup or comprehension support for ASL video environments.

Fortunately, recent advances in sign segmentation [24, 27, 32] and recognition [8, 41] now make it possible to automate this supportive workflow. In particular, segmentation [24] has the potential to provide learners with video segments and the closest matching signs for each segment.

We will demonstrate an automatic, end-to-end ASL video comprehension prototype system. Building on prior studies that employed a Wizard-of-Oz method to investigate design parameters [12], this tool was originally developed and released for research purposes as part of a study on AI-based ASL video comprehension [17][2]. The system incorporates automatic sign language segmentation functionality, and its underlying AI pipeline combines state-of-the-art sign language segmentation with a fine-tuned recognition model that operates on the segmented outputs. The goal of this demonstration is to showcase how automatic segmentation and sign lookup can support learners and to gather informal feedback on the interaction design and performance of the underlying systems from ASSETS attendees, including d/Deaf, Hard of Hearing, and hearing signers and learners.

2 Related Work

2.1 Isolated Sign Language Recognition

Early sign recognition used handcrafted features and small datasets [23, 40] and relied on SVMs or HMMs [30]. The shift toward deep learning introduced 2D-CNNs and RNNs, which were gradually replaced by 3D-CNNs and Transformers capturing spatiotemporal dependencies [24, 38]. Concurrently, training corpora evolved into large-scale datasets like *MS-ASL* [15], *WLASL* [35], and *ASL Citizen* [8], enabling highly generalizable architectures like the Sign Pose Transformer [4]. However, these models mostly focus on signs in their isolated "citation form." To bridge this, Zhou et al. [41] recently coupled Graph Convolutional Networks (GCNs) with temporal boundary detection, improving accuracy for pre-segmented signs.

2.2 Sign Language Segmentation

Continuous Sign Language Recognition (CSLR) identifies successive signs without prior boundary knowledge [2], but remains challenging due to coarticulation [29]. Even with emerging foundation models (e.g., SignGemma [11], SignX [10]), learners heavily benefit from analyzing shorter segments. Thus, sign segmentation—breaking continuous videos into individual glosses—is critical. Early methods used pseudo-labelling and changepoint detection to refine boundaries [27]. Recently, Moryossef et al. [24] proposed a *linguistically motivated segmentation* framework using a BIO-tagging scheme, optical-flow prosodic features, and 3D hand normalization to detect precise boundaries, yielding the most consistent cross-linguistic results [32].

This demonstration integrates these advances by fine-tuning a recognition model adapted for both citation forms and automated segments [8, 41] using robust datasets [8, 25, 26], while leveraging the linguistically motivated pipeline [24] for our user interface.

[2]Prototype system: https://github.com/chaelin0722/asl-video-lookup-tool

Table 1: Recognition performance of the fine-tuned ST-GCN model on the ASLLRP dataset [25] (mean ± standard deviation).

Top-1	Top-5	Top-10	Top-20
0.33 ± 0.30	0.44 ± 0.38	0.48 ± 0.40	0.51 ± 0.42

3 AI Pipeline Architecture

We describe the processing pipeline in Figure 2 and present an overview in this section. For further details, please refer to prior work and software documentation on the system components [8, 17, 18, 24]. Notably, Kim et al. [18] provide a summary of both the AI pipeline and the interface design.

For *automatic segmentation*, the system uses the linguistically motivated segmentation model proposed in [24]. This model adopts a BIO-tagging scheme (Beginning–Inside–Outside) to identify the start and end points of each sign in continuous signing. The approach is grounded in linguistic cues such as prosody, motion, and handshape changes, and jointly models both sign- and phrase-level boundaries using pose-based features extracted from the signer's body and hands. Because it relies on motion and structural patterns rather than language-specific gloss annotations, the model is expected to generalize well across different sign languages. For the *isolated sign recognition* model, the system uses the Spatial-Temporal Graph Convolutional Network (ST-GCN) architecture [8]. ST-GCN is a pose-based model that provides competitive performance relative to other state-of-the-art models while enabling faster inference, making it well suited for integration into a real-time demonstration. The model, initialized with pretrained weights from the ASL Citizen dataset [8], is fine-tuned on a dataset of 991 glosses for 200 epochs using Stochastic Gradient Descent (SGD) [28] with a learning rate of 1×10^{-1}. The dataset exhibits substantial class imbalance, with gloss instances ranging from 1 to 289 per class. To address this imbalance, the training procedure employs a weighted random sampler that assigns higher sampling probabilities to underrepresented glosses [18].

Following the overall model pipeline described in [8], the system extracts 27 keypoints from the hands and face using MediaPipe Holistic [22], then center-scales and normalizes them. Data augmentation techniques, including random rotation and shear transformations on pose sequences, are applied as described in the original paper [8]. For optimization, the system uses cross-entropy loss with a cosine annealing learning rate scheduler [21] (T_{max} = 150). The encoder is initialized with pretrained weights, and the classifier is reinitialized using Xavier initialization [7]. We evaluate the fine-tuned recognition model on the ASLLRP dataset [25]. Table 1 summarizes recognition performance.

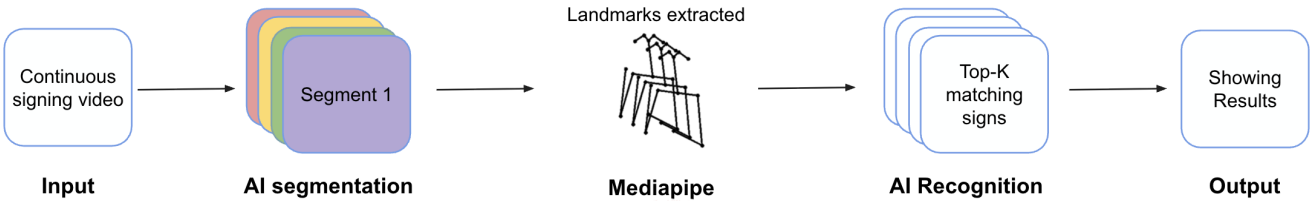


Figure 2: System architecture for automated ASL video comprehension. The system segments a continuous signing video into individual signs, extracts body and hand landmarks using MediaPipe, identifies the top-K matching signs, and presents the results to support learners' comprehension.

4 User Interface and Demo Plan

The demonstration will run on a 14-inch MacBook Pro with an M4 chip, which provides sufficient GPU compute power to support our models. We will provide attendees who may have a range of experience with ASL with a selection of sample videos at four different levels, pre-evaluated to ensure reasonable coverage of signs, which they can upload on a separate tablet. Next, we will familiarize them with the different components of the user interface with the help of the labeled diagrams in the accompanying poster.

As shown in Figure 3, the web-based system consists of five main components: a "`video player screen`", "`transcription panel`", "`search results panel`", "`video timeline`", and "`span selection functionality`". To support additional interactions implemented in the codebase [17], the system integrates four user-interface elements: a "`Segment and Search`" button, a "`Search`" button, and "`Upload Video`" and "`Recording`" buttons.

We will then encourage attendees to use the "`Segment and Search`" button, which allows them to trigger automatic segmentation and recognition on the segmentation output. After selecting a span on the video timeline using the adjustable playhead's start and end markers, attendees can trigger the segmentation system to divide the selection into individual signs. This selection can include the entire duration of the video or a smaller segment that the user wants to analyze, so we will allow attendees to explore freely.

Once attendees click "`Segment and Search`" and processing is complete, a confirmation message will appear: "`Processing done, showing 20 results between selected start time–end time.`" Each automatically detected sub-segment is displayed in the navigation bar in chronological order, and selecting a sub-segment highlights the corresponding region on the timeline. This will allow attendees to quickly assess recognition quality and compare retrieved signs to their target signs without leaving the video context. The retrieved results are shown in the "`Search Results`" panel, where each sign is represented as a looping GIF along with a "Gloss" label[3] and a confidence score. Confidence levels are generated based on the system's confidence and are grouped into three ranges: Low (0–33%), Medium (34–66%), and High (67–100%). Finally, we will give attendees an opportunity to provide informal feedback through a survey and engage in discussion about future directions.

[3]An ASL gloss is a written label, typically an English word in capital letters, used to represent an ASL sign. Glossing serves as an interlinear notation system for representing ASL's visual-spatial grammar and structure [34].

5 Conference Discussion Opportunities and Future Directions

There are several opportunities to improve the system's AI models and interface, which we hope to discuss with ASSETS attendees. We outline some of these discussion points and future directions below.

As prior interdisciplinary workshops by Bragg et al. [5] and others have shown, the availability of well-annotated large datasets with diverse signers remains one of the biggest constraints in sign language understanding research. Even when such datasets are disseminated, they may not be publicly available or sufficiently representative of the diversity of signing practices. In the context of our system design, there is also a need for more annotated continuous-signing video data with sign-level annotations, which remains limited. Recent datasets, such as YouTube-ASL [37], are promising but will require significant effort from future researchers to annotate large corpora of videos using English captions.

As noted in the original work [24], the segmentation model was initially trained on German Sign Language (Deutsche Gebärdensprache) data rather than ASL data. Fortunately, recent methods, including the approach used in prior work [24], leverage centroid representations across multiple examples to stabilize performance [9] and can still achieve reasonable performance across different sign languages. However, future researchers could further improve segmentation by leveraging labeled continuous ASL datasets [26], or comparable datasets for other sign languages of interest, to fine-tune sign language-specific segmentation architectures. Future work can also investigate the use of large vision language models, e.g., Google Gemma [36] or Qwen2-VL [39], with prompts for automatic segmentation tasks.

Future researchers can also implement several UI enhancements to further lower the cognitive workload for independent learners. The current system performs well with a single signer. However, many videos online include multiple signers, such as conversations between two signers, or present sign language in only a small portion of the screen, such as through a sign language interpreter. This creates another opportunity to explore the design of *spatial* video segmentation, as opposed to only *temporal* segmentation. Future researchers can investigate this interaction.

Another limitation of the current interface is that users cannot visualize the detected segments within the timeline. Future work could provide this visualization to enable users to evaluate the detected segments based on their own understanding and adjust the segment boundaries as needed. Although the exact design should be

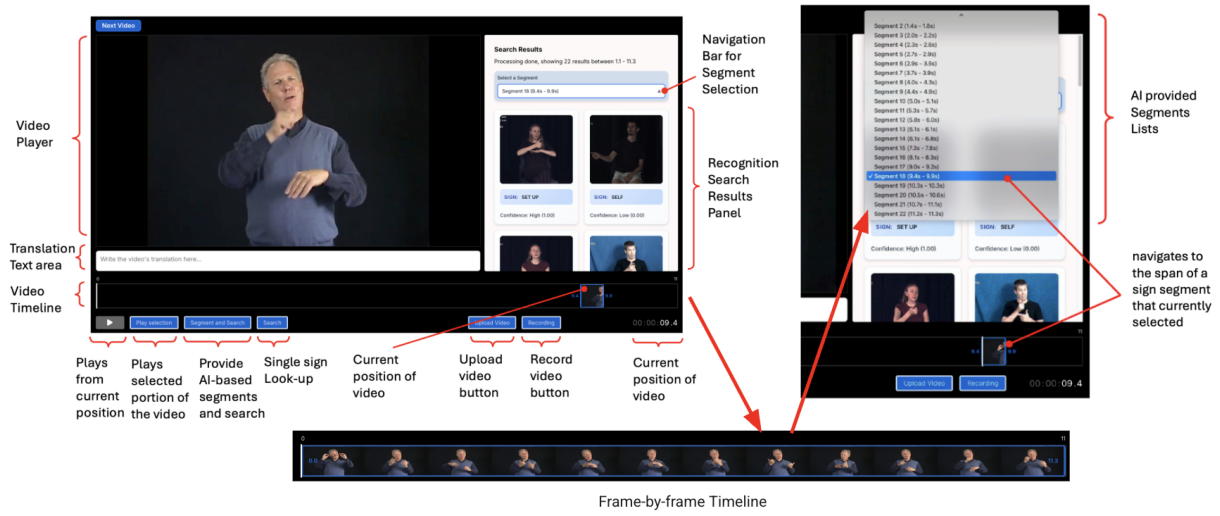


Figure 3: A scenario-based walkthrough of the learner interface. Users upload a signing video and activate "Segment & Search." The system automatically segments continuous signing into individual sign units and displays candidate recognition results for each segment in the results panel.

investigated, one idea is that the timeline bar could also be indexed with detected sign boundaries. By hovering over specific segments on the timeline, users could trigger immediate pop-ups displaying the top-matching glosses and confidence metrics, as proposed in prior research [14]. We hope the demo will surface similarly creative ideas for future investigation.

6 Conclusion

We plan to demonstrate a fully automated, end-to-end ASL video comprehension prototype that streamlines vocabulary lookup for sign language learners. The system utilizes an AI pipeline that automatically segments continuous signing videos and retrieves the closest matching sign glosses using an ST-GCN recognition model. The system provides top-k matching gloss-labeled GIFs directly within the learner's video-viewing environment. We hope to encourage researchers and practitioners to build such tools to support autonomous ASL learning and to discuss future directions.

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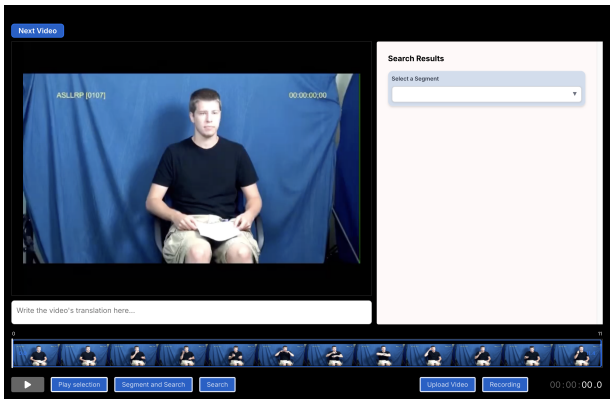
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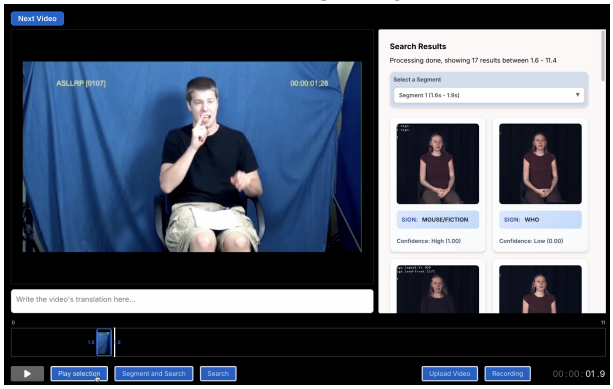
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A The interface before and after executing a search



(a) Prior to AI processing



(b) After AI processing

Figure 4: System interface states (a) before and (b) after executing the AI pipeline.